Assignment 1

① Given

Period: 4 years

Requirements

GPU: 8 GPUs

Compute Resources: Each server: → 32 core server
→ 256 GB RAM
→ 2TB SSD

Cloud: AWS p4d.24xlarge instances
→ 8 GPUs
→ 96 vCPUs
→ 1.1 TB RAM
→ 8 TB NVMe SSD

Usuage Patterns: → (8 hrs/day, 22 days/month) OR (176 hrs/month)

Data Transfer: 5TB of outbound traffic per month in the clould scenario.

On-Premise Server Cost List

Each server
↳ 4 GPUs, 32-core, SSD, networking.
↳ $65,800

Power, Cooling, IT Staff
↳ $5,000/year per server.

Cloud-Based Solution Cost List

AWS p4d.24xlarge
↳ $32.77/hour
(Locked in for 4 years)

Data Transfer
↳ 5TB/month.
↳ $0.09/GB.

② TCO of On-Prem

$(2 \times 65,800) + (5,000 \times 4 \times 2)$
$= \$171,600$

③ TCO of Cloud

$(32.77 \times 176 \times 4 \times 12) + (0.09 \times 5 \times 1000 \times 4 \times 12)$
$= 276,840.96 + 21,600$
$= \$298,440.96$

Therefore: TCO of on-prem is \$171,600.

TCO of cloud is \$298,440.96